

Analyzing Credit Risk with the Lending Club Dataset

1.1 Introduction to the Peer-to-Peer (P2P) Lending Model

Before diving into data analysis, it is crucial for students to understand the business model that generates the data. The dataset comes from Lending Club, a prominent player in the peer-to-peer (P2P) lending industry.¹ The P2P model functions as a financial marketplace that directly connects individual borrowers with individual or institutional investors, effectively bypassing traditional banking intermediaries.¹

In this ecosystem, Lending Club acts as the facilitator and servicer. The process begins when a potential borrower applies for an unsecured personal loan, which can range from \$1,000 to \$40,000.¹ Lending Club then performs a critical function: risk assessment. Based on the applicant's credit history, FICO score, debt-to-income ratio, and other financial indicators, Lending Club determines the borrower's creditworthiness. If approved, the loan is assigned a credit grade (from A to G) and a corresponding interest rate.¹ This grade and rate are designed to reflect the level of risk associated with the borrower; lower-risk borrowers receive more favorable rates, while higher-risk borrowers are assigned higher rates to compensate investors for the increased probability of default.⁴

Once a loan is graded and listed on the platform, investors can browse the available loans and choose which ones to fund, typically in small increments of as little as \$25 per loan.¹ This diversification allows investors to spread their risk across a large portfolio of loans. As borrowers make their monthly payments, the principal and interest (less a service fee for Lending Club) are passed on to the investors.²

The fundamental business challenge for Lending Club, and the central theme of this analytical curriculum, is the accurate pricing of risk. The platform must set interest rates that are low enough to attract qualified borrowers away from traditional banks, yet high enough to provide a compelling return for investors that justifies the risk of default.⁵ This delicate balance is the key to the platform's profitability and sustainability.

1.2 The Lending Club Dataset: Your Analytical Sandbox

The Lending Club dataset is an ideal resource for teaching financial analysis. It is a large, publicly available collection of real-world data containing detailed information on hundreds of thousands of loans issued through the platform.² This dataset provides a practical environment for students to apply their finance knowledge and develop data visualization skills.

Dataset Access

The most comprehensive and well-maintained version of this dataset is available on Kaggle. It is recommended to use this source as it is extensive and includes documentation on the data's provenance.⁷

- **Direct Download Link:**

<https://www.kaggle.com/datasets/wordsforthewise/lending-club>⁷

Data Structure and Scope

The dataset is robust, typically containing around 75 variables (columns) for each loan (row).² These variables cover a wide spectrum of information, including the loan's characteristics (amount, term, interest rate), the borrower's financial profile (income, credit score, employment history), and the loan's performance over time (payment history, current status).

A critical point to emphasize to students is the inherent selection bias in this data. The dataset represents only *accepted and funded loans*. It does not include information on loan applications that were rejected by Lending Club or loan offers that were not accepted by the borrower.⁶ This means any analysis performed is on a pre-screened population of borrowers deemed creditworthy enough to receive funding. Understanding this context is vital for drawing accurate conclusions and avoiding generalizations about the entire population of loan applicants.

1.3 Key Data Dictionary for Tableau

To prevent students from being overwhelmed by the sheer number of variables, the curriculum should initially focus on a curated set of the most impactful fields for credit risk analysis. The table below organizes these essential variables into logical categories, providing a structured mental model for understanding the credit assessment process.

Category	Field Name	Data Type	Description (Simplified for Teaching)	Relevant Snippets
Loan Characteristics	loan_amnt	Numeric	The total amount of money the borrower applied for.	2
	term	String	The duration of the loan, either	10

			"36 months" or "60 months".	
	int_rate	Numer ic	The interest rate charged on the loan. A key indicator of risk.	2
	installment	Numer ic	The fixed monthly payment the borrower must make.	2
	grade / sub_grade	String	Lending Club's internal risk rating (A-G). Higher grade = higher risk.	2
	purpose	String	The reason the borrower is taking the loan (e.g., 'debt_consolidation').	2

	issue_d	Date	The month and year the loan was funded.	9
Borrower Profile	fico_range_low / high	Numeric	The borrower's FICO credit score at the time of application.	2
	dti	Numeric	Debt-to-Income Ratio. A key measure of a borrower's ability to repay.	2
	annual_inc	Numeric	The borrower's self-reported annual income.	2
	home_ownership	String	The borrower's housing situation (RENT, OWN, MORTGAGE).	2
	revol_util	Numeric	Revolving Credit Utilization. The percentage of	5

			available credit being used.	
	emp_length	String	The borrower's length of employment in years.	2
	verification_status	String	Indicates if the borrower's income was verified by Lending Club.	2
Loan Outcome	loan_status	String	The current status of the loan. The primary target for our analysis.	12
	total_pymnt	Numeric	The total amount of money paid back by the borrower so far.	11

	recoveries	Numer ic	Any funds recovered by a collection agency after a loan has charged off.	9
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Section 2: The Language of Lending: Essential Financial Concepts for Your Students

To effectively analyze the Lending Club data, students must be fluent in the language of credit and risk. This section bridges the user's domain knowledge gap by deconstructing the key financial concepts embedded within the dataset. Each concept is explained in clear, accessible terms and directly linked to the relevant data fields students will use in Tableau.

2.1 Loan Performance: The Lifecycle of a Loan

The most critical variable for risk analysis is `loan_status`, as it describes the ultimate outcome of the lending decision. Understanding its various states is fundamental.

- **Fully Paid:** This is the ideal outcome for both the borrower and the investor. The borrower has successfully met all their obligations, paying back the entire loan principal plus all accrued interest over the loan term.¹⁴
- **Current:** This status indicates that the loan is active and in good standing. The borrower is up-to-date with their monthly payments, and the loan is performing as expected.
- **Default vs. Charged Off:** These terms are often used interchangeably in casual conversation, but in finance and accounting, they represent distinct stages of loan delinquency. A

loan typically enters a state of **Default** after a borrower has missed payments for a specified period (e.g., 90-120 days). This is the point where the lender considers the loan agreement to be broken. Following default, if the debt remains uncollected after a further period (often 180 days), the lender will take an accounting action known as a **Charge-Off**.¹⁴ This means the lender declares the debt to be uncollectible and writes it off as a loss on its income statement.¹⁴

It is essential to teach students that a charge-off does not signify debt forgiveness. The borrower is still legally obligated to repay the debt.¹⁵ The lender can continue to pursue collection efforts or, more commonly, sell the charged-off debt to a third-party collection agency for a fraction of its face value.¹⁴

This distinction leads to a more sophisticated analysis of financial loss. Simply counting the number of "Charged Off" loans or summing their original amounts (`loan_amnt`) overstates the actual financial loss to the investor. The dataset contains a field named `recoveries`, which tracks any funds collected after a loan has been charged off.⁹ Furthermore, borrowers may have made substantial payments before defaulting, which are captured in the

`total_pymnt` field. Therefore, a more precise measure of the actual financial loss on a defaulted loan can be calculated. This provides an excellent opportunity to teach students how to derive meaningful business metrics from raw data by creating a calculated field in Tableau for **Net Loss**:

Net Loss=loan_amnt-total_pymnt-recoveries

Analyzing this calculated metric provides a much more accurate picture of the financial impact of defaults than simply looking at the `loan_status` field in isolation.

2.2 The Borrower's Profile: Deconstructing Creditworthiness

Before a loan is ever issued, lenders perform a detailed assessment of the borrower's financial health. The dataset contains several key metrics used in this underwriting process. These are the "input" variables that help predict the loan's outcome.

2.2.1 The FICO Score

The FICO score is a cornerstone of consumer lending in the United States. It is a three-digit number, typically ranging from 300 to 850, that summarizes an individual's credit risk at a specific point in time.¹⁹ Lenders use this score as a quick and consistent way to assess the likelihood that a borrower will repay their debts as agreed.¹⁹ A higher score indicates lower risk. The score is calculated from information in a consumer's credit report and is based on five key factors with varying weights¹⁹:

1. **Payment History (35%):** The most important factor. It reflects whether the individual has paid past credit accounts on time.
2. **Amounts Owed (30%):** This considers the total amount of debt, the number of accounts with balances, and the credit utilization ratio.
3. **Length of Credit History (15%):** A longer history of responsible credit management is generally favorable.
4. **New Credit (10%):** This looks at recent credit inquiries and newly opened accounts. Opening many new accounts in a short period can be a sign of risk.
5. **Credit Mix (10%):** This considers the variety of credit accounts, such as credit cards, mortgages, and installment loans.

Students should be familiar with the general FICO score ranges to interpret the `fico_range_low` and `fico_range_high` fields in the dataset ²⁰.

- **Exceptional:** 800+
- **Very Good:** 740–799
- **Good:** 670–739
- **Fair:** 580–669
- **Poor:** < 580

2.2.2 The Debt-to-Income (DTI) Ratio

The Debt-to-Income (DTI) ratio is another critical metric for lenders. It measures a borrower's ability to manage their monthly debt payments given their income.²³ It is calculated as the percentage of a borrower's gross monthly income (income before taxes) that goes towards servicing their existing recurring debt payments (e.g., mortgage, auto loans, student loans, minimum credit card payments).²⁴

The formula is straightforward:

$$\text{DTI} = \frac{\text{Gross Monthly Income}}{\text{Total Monthly Debt Payments}} \times 100$$

For example, if an individual has \$2,000 in monthly debt payments and a gross monthly income of \$6,000, their DTI is 33.3%.²⁵ Lenders use DTI to gauge whether a borrower can afford to take on new debt. While standards vary, some general industry benchmarks can provide context for the

`dti` field ²⁶:

- **≤36%:** Generally considered a good DTI, indicating the borrower has a manageable debt load.
- **37%–43%:** Manageable, but indicates some risk. Lenders may require additional eligibility factors.

- **>43%:** Often considered too high, suggesting the borrower may be overextended and could have trouble making payments on a new loan.

2.2.3 Revolving Balance and Credit Utilization (**revol_util**)

It is important for students to distinguish between different types of debt. **Installment loans** (like auto loans or mortgages) have a fixed number of equal payments. **Revolving credit** (like credit cards or lines of credit) allows a borrower to borrow and repay funds up to a set credit limit on an ongoing basis.³⁰

The **revol_util** field refers to the borrower's **revolving credit utilization ratio**. This metric compares the total outstanding balance on all their revolving accounts to their total available credit limit.¹³ For example, if a person has a total credit limit of \$10,000 across all their credit cards and currently has a combined balance of \$4,000, their utilization ratio is

40%.

This ratio is a significant component of the "Amounts Owed" category in the FICO score calculation. A high utilization rate can signal to lenders that a borrower is overly reliant on credit and may be at risk of financial distress. A common rule of thumb is to keep credit utilization below 30% to maintain a healthy credit score.³⁰

2.3 The Lender's Assessment: Understanding **grade** and **sub_grade**

The **grade** and **sub_grade** fields are not raw data from the borrower but are instead a proprietary output from Lending Club's own internal risk assessment model.³ The platform's algorithm analyzes all the available borrower profile data—FICO score, DTI, income,

employment history, etc.—to produce a single, consolidated risk rating from A (lowest risk) to G (highest risk).¹

This grade directly determines the interest rate offered to the borrower. A borrower with an 'A' grade represents a very low risk of default and will be offered a low, competitive interest rate. Conversely, a borrower with an 'F' grade represents a significantly higher risk, and the interest rate will be much higher to compensate investors for taking on that risk.¹ Analysis of the data confirms this relationship: default rates steadily increase as the loan grades worsen from A to G.⁴

This presents a powerful teaching opportunity. The **grade** can be viewed as a pre-packaged prediction from a sophisticated, "black box" model. For students, it serves as an excellent benchmark. The initial analysis will confirm that the grade is a strong predictor of default. However, a more advanced analysis, and a great challenge for students, is to use Tableau to investigate which specific borrower characteristics are the strongest drivers of default *within* a particular grade. For example, among all 'C' grade loans, do those with a **purpose** of 'small_business' default more often than those for 'debt_consolidation'? This type of granular analysis mimics how a real-world data science team would work to refine and improve upon an existing risk model, moving beyond simple validation to generate new, actionable insights.

Section 3: Curriculum Module 1 - The 30,000-Foot View: A Portfolio Overview Dashboard

This foundational module serves as an accessible entry point for students. The goal is to familiarize them with the dataset and fundamental Tableau functionalities while answering high-level business questions. The resulting dashboard provides an

executive-level summary of the loan portfolio's health and composition.

3.1 Business Objective

To create a single, consolidated dashboard that provides a high-level, at-a-glance overview of the entire Lending Club loan portfolio. This dashboard is intended for senior management to quickly understand key business trends and the overall scale of operations.

3.2 Key Questions to Answer

This dashboard should answer the following fundamental business questions:

1. How has our loan issuance, measured in both total dollar amount and the number of loans, grown over the years?
2. Which states represent our largest markets in terms of loan volume and value?
3. What are the most common reasons people request loans from our platform?
4. What is the distribution of our loan portfolio between shorter (36-month) and longer (60-month) terms?

3.3 Tableau Skills to Teach

This module focuses on core Tableau skills that form the building blocks for all subsequent analysis:

- **Data Connection:** Connecting to a text file (CSV) and understanding the data source pane.
- **Basic Aggregations:** Creating Key Performance Indicators (KPIs) using fundamental aggregations like `SUM(loan_amnt)` and `COUNTD(id)`.
- **Date Fields and Time-Series Analysis:** Working with date fields (`issue_d`) to build time-series line charts and understanding date hierarchies (year, quarter, month).

- **Geographic Visualization:** Generating geographic maps (filled maps) by assigning a geographic role to the `addr_state` field.
- **Categorical Analysis:** Building simple bar charts and pie charts to visualize the distribution of categorical data like `purpose` and `term`.
- **Dashboard Assembly:** Combining multiple worksheets into a single, interactive dashboard using layout containers to organize the visualizations logically.

3.4 Dashboard Blueprint

The final dashboard should be organized for clarity and immediate impact, following a logical flow from overall trends to specific breakdowns.

- **Top Row (KPIs):** A series of "Big Ass Numbers" (BANs) or scorecards that display the most critical top-line metrics.
 - **Total Loan Amount:** `SUM([loan_amnt])` formatted as currency.
 - **Total # of Loans:** `COUNTD([id])` formatted as a number.
 - **Average Interest Rate:** `AVG([int_rate])` formatted as a percentage.
 - **Average Loan Amount:** `AVG([loan_amnt])` formatted as currency.
- **Main Left Panel (Time-Series Trend):** A dual-axis line chart to visualize growth over time.
 - **Columns:** Continuous `MONTH(issue_d)`.
 - **Rows (Axis 1):** `SUM(loan_amnt)` displayed as a line or area chart.
 - **Rows (Axis 2):** `COUNTD(id)` displayed as a line chart.
 - **Title:** "Loan Issuance Growth Over Time". This chart will clearly show the trajectory of the business.³⁴
- **Main Right Panel, Top (Geographic Distribution):** A filled map of the United States.

- **Marks:** Map type set to "Filled Map".
- **Color:** SUM(loan_amnt).
- **Tooltip:** Should display the state, total loan amount, and number of loans.
- **Title:** "Loan Concentration by State". This visualization immediately highlights key markets.³⁶
- **Main Right Panel, Bottom (Loan Purpose Breakdown):** A horizontal bar chart, sorted for easy comparison.
 - **Rows:** purpose.
 - **Columns:** COUNTD(id).
 - **Sort:** Descending by the count of loans.
 - **Title:** "Top Loan Purposes". This chart reveals the primary use cases for the platform, with 'debt_consolidation' typically being the most prominent.³⁵

Section 4: Curriculum Module 2 - Profiling the Borrower: A Risk Analysis Dashboard

This module transitions from a high-level overview to a more granular analysis of the borrowers themselves. The focus is on understanding the financial characteristics of the borrower base and exploring the fundamental relationship between borrower risk and the loan terms they receive. This dashboard introduces more sophisticated chart types and the concept of statistical correlation.

4.1 Business Objective

To analyze the financial profile of the typical Lending Club borrower and to visually demonstrate how their creditworthiness, as measured by key financial metrics, directly influences the interest rates assigned by the platform. This dashboard helps answer the question: "Are we pricing risk appropriately?"

4.2 Key Questions to Answer

This analysis will delve into the characteristics that define a borrower's risk level:

1. What is the distribution of borrower FICO scores and Debt-to-Income (DTI) ratios across our entire portfolio? Are we lending primarily to prime or subprime borrowers?
2. Is there a clear, quantifiable relationship between a borrower's FICO score and the interest rate they are offered?
3. How does Lending Club's internal loan `grade` map to objective risk metrics like FICO scores and the resulting interest rates?
4. Does the verification status of a borrower's income correlate with different risk profiles or loan terms?

4.3 Tableau Skills to Teach

This module builds upon the fundamentals by introducing more advanced analytical techniques and chart types:

- **Distribution Analysis:** Creating histograms to visualize the frequency distribution of continuous measures like `fico_range_low` and `dti`. This involves using bins.
- **Correlation Analysis:** Building scatter plots to explore the relationship between two continuous variables (e.g., FICO score and interest rate).
- **Statistical Tools:** Adding trend lines to scatter plots and interpreting their slope and R-squared values to understand the strength and direction of a correlation.
- **Calculated Fields:** Using simple calculated fields for aggregations like `AVG(int_rate)` and `AVG(fico_range_low)`.
- **Advanced Visual Encoding:** Using color, shape, and size marks to encode additional categorical (`grade`, `verification_status`) and continuous dimensions onto a single visualization.

- **Comparative Analysis:** Creating box-and-whisker plots to compare the distribution of a measure across different categories (e.g., comparing the distribution of FICO scores for each loan grade).
- **Interactivity:** Introducing dashboard filters to allow users to slice the data and explore these relationships for different borrower segments.

4.4 Dashboard Blueprint

The dashboard should be designed to tell a clear story about risk and pricing, with each component contributing to the central narrative.

- **Top Left (Credit Quality Distribution):** A histogram built from the `fico_range_low` field.
 - **Columns:** `fico_range_low` (binned).
 - **Rows:** `COUNTD(id)`.
 - **Title:** "Distribution of Borrower FICO Scores". This chart provides an immediate sense of the overall credit quality of the portfolio.⁵
- **Top Right (Debt Burden Distribution):** A histogram built from the `dti` field.
 - **Columns:** `dti` (binned).
 - **Rows:** `COUNTD(id)`.
 - **Title:** "Distribution of Borrower Debt-to-Income Ratios". This visualizes the debt load of the borrower base.
- **Bottom Panel (Core Risk-Pricing Model):** A large, detailed scatter plot that serves as the centerpiece of the dashboard.
 - **Columns (X-axis):** `AVG(fico_range_low)`.
 - **Rows (Y-axis):** `AVG(int_rate)`.
 - **Detail:** `sub_grade`. Each mark on the plot represents a specific sub-grade (e.g., A1, A2, B1, B2).
 - **Color:** `grade`. This will visually group the sub-grades into their parent grade categories.

- **Trend Line:** Add a linear trend line. This will visually demonstrate a strong negative correlation: as the average FICO score increases, the average interest rate decreases.⁵ This is the visual proof of a rational risk-pricing model.
- **Title:** "Interest Rate vs. FICO Score by Loan Grade".
- **Interactivity (Filters):** Add several filters to the right side of the dashboard to encourage exploration.
 - **Filter 1:** `home_ownership` (multi-select dropdown).
 - **Filter 2:** `purpose` (multi-select dropdown).
 - **Filter 3:** `emp_length` (multi-select dropdown).
 - These filters will allow students to investigate whether the core relationship between FICO and interest rate holds true for different borrower segments, such as renters applying for small business loans.

Section 5: Curriculum Module 3 - When Loans Go Bad: A Default Analysis Dashboard

This module is the analytical core of the curriculum for a finance-focused audience. It moves from profiling and pricing to performance, focusing directly on credit risk by analyzing the factors that drive loan defaults. This is where students will create the most critical business metrics from scratch.

5.1 Business Objective

To identify the key borrower characteristics and loan segments that are most strongly associated with loan defaults ("Charged Off" status). The insights from this dashboard are designed to inform risk management strategies, potentially leading to adjustments in underwriting criteria or pricing for high-risk segments.

5.2 Key Questions to Answer

This dashboard is built to answer the most pressing questions for a credit risk manager:

1. What is the overall default rate for the entire loan portfolio?
2. How does the default rate vary across different loan **grades**?
Does our internal grading system effectively predict defaults?
3. Are certain loan **purposes** (e.g., small business, wedding) or **home_ownership** statuses (e.g., rent vs. mortgage) inherently riskier than others?
4. What is the total financial impact, in terms of the dollar amount charged off, of defaults originating from different segments?

5.3 Tableau Skills to Teach

This module places a strong emphasis on creating powerful calculated fields to transform raw data into actionable metrics.

- **Advanced Calculated Fields:** This is the central skill of this module. Students must learn to create logical and conditional calculations to define and measure risk.
 - Is Default?: A binary flag to identify defaulted loans.
`IF [loan_status] = "Charged Off" THEN 1 ELSE 0 END`
 - Default Rate: The primary risk metric, calculated as a percentage.
`SUM() / COUNTD([id])`
 - Total Charged Off Amount: A measure of the financial impact of defaults.
`SUM(IF = 1 THEN [loan_amnt] ELSE 0 END)`
- **Comparative Bar Charts:** Creating bar charts that use the new **Default Rate** calculation to compare risk across different categorical dimensions (**grade**, **purpose**, **home_ownership**).
- **Magnitude Visualizations (Tree Maps):** Using tree maps to effectively visualize the magnitude of a measure, such as the **Total Charged Off Amount**, across many categories simultaneously.

- **Dashboard Interactivity:** Implementing dashboard actions (e.g., Filter Actions) to create a more dynamic and exploratory user experience. For instance, clicking on a specific **grade** in one chart could filter all other charts on the dashboard to show data only for that grade.

5.4 Dashboard Blueprint

The layout should guide the user from the overall default rate to specific high-risk segments, combining both the probability (rate) and impact (amount) of defaults.

- **Top Row (KPIs):** Key summary metrics that provide an immediate snapshot of portfolio risk.
 - **Overall Default Rate:** The calculated **Default Rate** for the entire dataset, displayed prominently as a percentage.
 - **Total Charged Off Amount:** The calculated **Total Charged Off Amount**, formatted as currency.
- **Left Panel (Grade Performance):** A prominent vertical bar chart that validates the effectiveness of Lending Club's risk grading.
 - **Columns:** **grade** (sorted A to G).
 - **Rows:** **Default Rate** (formatted as a percentage).
 - **Title:** "Default Rate by Loan Grade". This chart should reveal a clear, incremental "stairstep" pattern, with the default rate increasing as the grade worsens. This visually confirms that the grading system is a strong predictor of default.⁴
- **Right Panel (Purpose Risk):** A horizontal bar chart to identify high-risk loan purposes.
 - **Rows:** **purpose**.
 - **Columns:** **Default Rate**.
 - **Sort:** Descending by **Default Rate**.
 - **Title:** "Default Rate by Loan Purpose". This chart will quickly highlight which categories, such as

'small_business', carry a disproportionately high risk of default compared to others like 'debt_consolidation'.³⁶

- **Bottom Panel (Geographic Risk Impact):** A tree map that combines two metrics to show both the location and financial severity of defaults.
 - **Marks:** Set to Tree Map.
 - **Size:** Total Charged Off Amount. The size of each rectangle will represent the total dollar amount lost to defaults in that state.
 - **Color:** Default Rate. The color of each rectangle will represent the default rate in that state (e.g., from green for low rates to red for high rates).
 - **Label:** addr_state.
 - **Title:** "Financial Impact of Defaults by State". This powerful visualization allows a manager to see, for example, that while California may have the highest total charged-off amount (due to its large volume), a smaller state might have a much higher and more concerning default rate.

Section 6: Curriculum Module 4 - From Analyst to Advisor: Building an Interactive Risk Scorecard

This capstone module is designed to elevate students from data reporters to data advisors. It challenges them to synthesize all the skills learned in the previous modules to build a dynamic, forward-looking, and interactive tool. The final output is not just a dashboard but a data-driven narrative that leads to a business recommendation.

6.1 Business Objective

To create an interactive risk scorecard that allows a loan officer, underwriter, or portfolio manager to perform "what-if" analysis on

specific borrower segments. This tool should enable users to define a segment based on multiple criteria and immediately see its historical risk and return profile, empowering them to make more informed, data-driven decisions. The module culminates in using Tableau's Story Points feature to present these findings as a compelling briefing.

6.2 Key Questions to Answer

This interactive tool is designed to answer dynamic, user-driven questions:

1. For a specific combination of borrower characteristics (e.g., Grade 'C' borrowers with a DTI between 20-30%, seeking a loan for 'debt_consolidation'), what is the historical default rate and the average interest rate charged?
2. How does this custom-defined segment's risk and return profile compare to the overall portfolio average?
3. What is the overarching story of risk and return in our portfolio, and which specific segments require strategic attention?

6.3 Tableau Skills to Teach

This module introduces some of the most powerful features in Tableau for creating advisory and exploratory analytics:

- **Parameters:** Creating user-controlled variables (parameters) for key dimensions like `Grade`, `Loan Purpose`, and for ranges of continuous measures like `DTI` and `FICO Score`. This is the core of "what-if" analysis.
- **Complex Calculated Fields with Parameters:** Writing more advanced calculations that incorporate parameters to dynamically filter the view. For example:
`[grade] = [Grade Parameter] AND [dti] >= AND [dti] <=`
- **Dynamic KPIs and BANs:** Creating worksheet titles and text boxes that update automatically based on parameter selections to display key metrics for the chosen segment.

- **Comparative Analytics with Reference Lines:** Adding portfolio-average reference lines to charts, allowing for instant visual comparison between the selected segment and the overall benchmark.
- **Data Storytelling with Story Points:** Guiding students on how to structure a narrative by combining their previously built dashboards and new annotations into a sequential story, moving from a broad overview to a specific, actionable recommendation.

6.4 Dashboard & Story Blueprint

The final project consists of two parts: the interactive dashboard and the narrative story built from it.

Dashboard: The Interactive Risk Scorecard

- **Left Control Panel:** A dedicated section of the dashboard containing all the user-facing controls.
 - **Parameters:**
 - A single-select dropdown for `Grade`.
 - A single-select dropdown for `Loan Purpose`.
 - Range sliders for `DTI` and `FICO Score`.
- **Main Panel (Dynamic KPIs):** A series of large, clear text-based visualizations that serve as the scorecard's output. These worksheets will be filtered by a calculated field that uses the parameters.
 - **"Segment Default Rate":** Displays the calculated default rate for the user-defined segment.
 - **"Segment Avg. Interest Rate":** Displays the average interest rate for the segment.
 - **"Number of Loans in Segment":** Displays the `COUNTD(id)` to give the user a sense of the sample size for their selection.
- **Comparison Charts:** Small, simple visualizations that provide immediate context.

- Bullet graphs or bar-in-bar charts comparing the "Segment Default Rate" to the "Overall Default Rate" and the "Segment Avg. Interest Rate" to the "Overall Avg. Interest Rate." These use reference lines to show the portfolio average.

Story: A Risk Committee Briefing

Students will create a new Story in Tableau and use their dashboards to present a concise, compelling analysis.

- **Story Point 1: The Big Picture.**
 - **Dashboard Used:** Module 1 (Portfolio Overview).
 - **Narrative:** "Our loan portfolio has experienced significant growth, reaching a total of \$X billion across Y funded loans. Our primary markets are concentrated in states like California, New York, and Texas."
- **Story Point 2: Our Pricing Model is Sound.**
 - **Dashboard Used:** Module 2 (Borrower Profile).
 - **Narrative:** "Our core business model effectively prices for risk. As this chart demonstrates, there is a strong, negative correlation between a borrower's FICO score and their assigned interest rate, confirming that we charge higher rates for higher-risk borrowers."
- **Story Point 3: Identifying a High-Risk Segment.**
 - **Dashboard Used:** Module 3 (Default Analysis).
 - **Narrative:** "However, a deeper analysis of loan performance reveals a key area of concern. Loans for the purpose of 'Small Business' have a default rate of Z%, which is significantly higher than our portfolio average and other major categories." ³⁶
- **Story Point 4: Drilling Down with the Scorecard.**
 - **Dashboard Used:** Module 4 (Interactive Scorecard).

- **Narrative:** "Using our interactive scorecard to isolate this segment, we can see the problem in greater detail. Even for 'C' grade borrowers—who are typically considered average risk—the default rate for small business loans is unacceptably high, without a commensurate increase in the average interest rate to compensate for this risk."
- **Story Point 5: Actionable Recommendation.**
 - **Dashboard Used:** Module 4 (Interactive Scorecard).
 - **Narrative: "Recommendation:** Based on this analysis, the risk committee should consider two actions: 1) Tighten the underwriting standards specifically for the 'Small Business' loan category, potentially requiring higher FICO scores or lower DTI ratios. 2) Re-evaluate our interest rate pricing model for this segment to ensure the rates charged adequately compensate investors for the elevated level of default risk."

This final module transforms the student from a passive analyst into an active participant in the business decision-making process, providing them with a powerful, portfolio-ready project.